

How life began

The very first life forms appeared on Earth around 3.5 billion years ago. Fossils preserved in rock help us chart the story of life from the first single-celled bacteria to the modern humans who roam Earth today.

DIVISION OF TIME

Earth's geological history can be divided into blocks of time. An era represents several hundred million years and is split into smaller periods. Earth is currently in the Quaternary Period of the Cenozoic Era.

-  **PRE-CAMBRIAN**
4.6 BYA–541 MYA
-  **PALAEZOIC ERA**
541–252 MYA
-  **MESOZOIC ERA**
252–65 MYA
-  **CENOZOIC ERA**
65 MYA–PRESENT DAY

BYA = Billion years ago
MYA = Million years ago

▶ 4.6 BYA–541 MYA

PRE-CAMBRIAN

This represents 80 per cent of total geological time. Volcanic activity on the new Earth produced water. Simple lifeforms appeared, and some produced oxygen.



VOLCANIC EARTH

▶ 541–485 MYA

CAMBRIAN

Many types of marine life evolved in the so-called "Cambrian explosion". They included molluscs, sponges, and animals with jointed legs (arthropods).



TRILOBITE

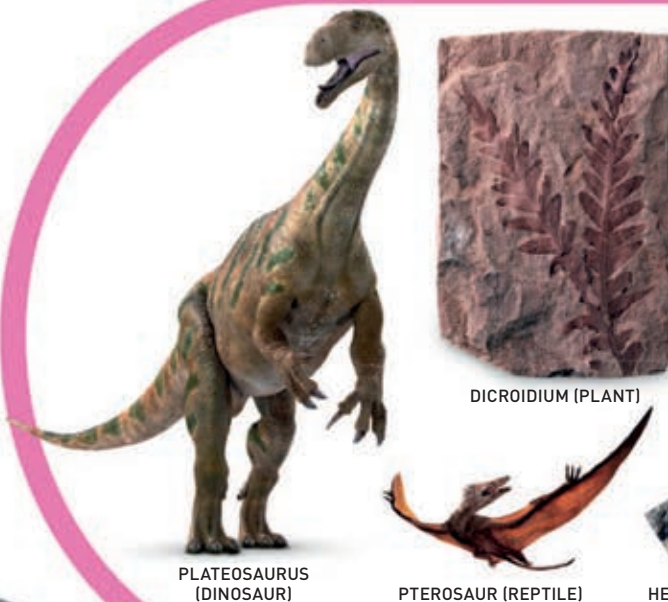


ECHINOCRINUS

252–201 MYA

TRIASSIC

The mass extinction of the late Permian meant that different types of life evolved. The first mammals, flying reptiles, and dinosaurs such as the long-necked *Plateosaurus* appeared.



PLATEOSAURUS
(DINOSAUR)

DICROIDIUM (PLANT)

PTEROSAUR (REPTILE)

AMMONITE

MORGANUCODON
(MAMMAL)

HEXAGONOCAULON (PLANT)

CRINOID (MARINE ANIMAL)

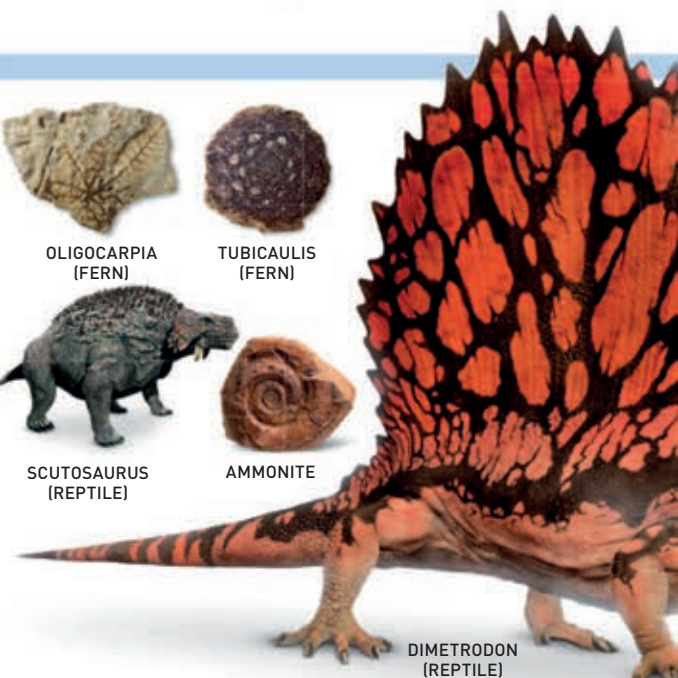


OLIGOCARPIA
(FERN)

TUBICAULIS
(FERN)

SCUTOSAURUS
(REPTILE)

AMMONITE



DIMETRODON
(REPTILE)

▶ 201–145 MYA

JURASSIC

Reptiles began to dominate the land and sea, and some took to the air. Many different dinosaurs roamed Earth, from giant plant-eaters to fierce predators.



PTERODACTYL (REPTILE)

EOCAECILIA (AMPHIBIAN)



DAKOSAURUS
(REPTILE)



WILLIAMSONIA
(PLANT)



KENTROSAURUS
(DINOSAUR)

▶ 145–65 MYA

CRETACEOUS

The climate was mostly warm, but cooled towards the end of the period. Flowering plants spread. The period ended with the mass extinction of the big dinosaurs.



EOMAIA (MAMMAL)

VEGAVIS (BIRD)



CONFUCIUSORNIS
(BIRD)



ARCHAEANTHUS
(PLANT)



ARCHELON
(TURTLE)



HOPLOPTERYX (FISH)



PROTOSTEGA (TURTLE)



SCAPHITES (AMMONITE)



CARNOTAURUS
(DINOSAUR)